Final Report

Problem Statement: Automatic Category Suggestions for Wikimedia Commons Images

Introduction and Problem Understanding

Wikimedia Commons is the shared media repository for Wikipedia and its sister projects, and it receives tens of thousands of new images every day. Volunteers organize files by assigning categories, which are the main way users find, search, and reuse media. Manual categorization cannot keep pace with the volume of uploads, so many images end up uncategorized or poorly categorized and are effectively invisible. This undermines the mission of Commons as an open and reusable repository.

The task is to build a system that suggests relevant categories for newly uploaded images. It assists human editors rather than replacing them: suggestions are reviewed before anything is applied. The system must handle objects, scenes, events and abstract concepts, generalize to unseen images, tolerate poor-quality photos, scale to Commons volumes, and improve from editor feedback.

The category space on Commons has millions of categories and a long tail where most categories have few examples. A fixed classifier would need retraining for every new category and cannot handle a category with zero examples. Images often have several categories, and stored labels are incomplete: editors usually assign only the most specific category and omit broader parents. These properties shaped our retrieval-based design.

Approach: Retrieval, Not Classification

We frame categorization as retrieval. Each image is converted into a 512-dimensional embedding with a pre-trained CLIP model (OpenCLIP ViT-B/32), which places images and text in a shared space. Categories are ranked using four complementary scorers, fused into one score, and cleaned using the category graph. Adding a new category only requires embedding its name, zero-shot matching works with no training examples, and a new image can join the kNN index without retraining.

System Architecture

Zero-shot scorer. A softmax over the similarity between the image embedding and the text embedding of "a photo of [category]". It needs no training and works for rare and new categories. **k-Nearest-Neighbour scorer.** The K=20 most similar indexed images vote for their categories, weighted by similarity. It also gives a built-in explanation: the neighbours themselves. **Trained head.** A linear layer with sigmoid outputs on the CLIP embedding, trained as a multi-label model with an asymmetric loss that copes with many negatives and incomplete labels. **Title/description scorer.** TF-IDF similarity between the title and description of the new file and the indexed files; the K=20 most similar files vote with their categories. It helps when a file name names the subject.

Fusion. A single logistic regression takes the four scores of an (image, category) pair and outputs one combined score. Candidates are the union of each scorer's top 30. Fusion is fitted on the validation split only, and it learns how much to trust each scorer (one set of weights shared across categories).

Hierarchy clean-up. We fetch the real Commons category graph, two levels of parents, through the MediaWiki API. Editors follow a "most-specific" rule, so in the demo a parent category is hidden only when a narrower category in the list scores at least the display cutoff. In evaluation we use lenient precision, where an ancestor of a true label is counted as acceptable.

Data and Evaluation

We downloaded images through the MediaWiki API from 41 seed categories (monuments, animals, scenes, events, objects and abstract patterns). After removing process categories (upload tools, license tags, dates) and keeping categories with at least 5 training images, the dataset has 3,354 images and 131 categories. Each image has a title, timestamp, license, description and categories.

We use a strict time-based split: the oldest 70% of images are used for training (2,348), the next 15% for validation (502, used to fit fusion), and the newest 15% for testing (500). This mimics deployment, where a system built today must handle tomorrow's uploads. Validation and test images whose cosine similarity to any training image is ≥ 0.97 are removed as near-duplicates. Because every image was found through a seed category, we also report *hard mode*, where all seed labels are hidden.

We report Precision@k, Recall@k and F1@k for $k = 1, 3, 5, 10$, and mAP. Because most images have only about two true labels, P@k for large k is naturally low, so we focus on P@1, R@5 and mAP. We break results down by category frequency (head, torso and tail are thirds of the categories by training frequency) and by category type, and test robustness to image damage. All numbers below are on the test split.

Results

Scorer	P@1	R@5	mAP	Hard P@1
Zero-shot	0.598	0.865	0.723	0.636
kNN	0.670	0.835	0.749	0.545
Trained head	0.696	0.880	0.779	0.455
Title/description	0.648	0.690	0.673	0.455
Fusion	**0.834**	**0.918**	**0.873**	**0.727**

Table 1: Test results (500 images, 131 categories). Hard mode hides all seed labels and is evaluated on only a small number of test images, so its values are rough.

Fusion gives the best overall ranking: it has the highest P@1, R@5 and mAP, and its strict P@1 is 0.138 above the best single scorer (the head, 0.696). Hard-mode values are lower than strict, which confirms that the seed label made the task easier.

Recall@5	Head	Torso	Tail
Zero-shot	0.861	**0.832**	**0.862**
Trained head	0.913	0.727	0.793
kNN	0.906	0.543	0.552
Title/description	0.781	0.412	0.207
Fusion	**0.960**	0.714	0.828

Table 2: Recall@5 by category frequency tier.

Fusion is not best on every slice. It is best on head categories (0.960), but on torso and tail categories the zero-shot scorer reaches higher recall (0.832 vs. 0.714 and 0.862 vs. 0.828). Zero-shot does not depend on training examples, while kNN and the text scorer fall sharply on rare categories. A likely reason for fusion's weaker tail is that the validation split holds few positives for rare categories; we did not test this. Taking the graph into account, strict P@5 is 0.190 and lenient P@5 (ancestors accepted) is 0.196.

Robustness and Efficiency

Test images after damage	P@5	R@5
Clean	0.190	0.918
Gaussian blur (radius 3)	0.184	0.892
Shrink 6× and stretch back	0.185	0.898
JPEG quality 10	0.189	0.918

Table 3: Fusion on damaged test images (titles still provided).

Recall falls by at most 2.6 points under blur and low resolution, and JPEG compression changes nothing. The title/description scorer does not use pixels, and titles were still given to it in this test, which helps. On a laptop CPU, one suggestion takes a median of 132 ms including the CLIP encoder, and batched processing reaches 34.2 images/s at batch size 32. The index of 3,354 images is 6.9 MB. Our kNN uses exact search, which is fine at this scale; at Commons scale the same scorer would move to an approximate index, which we have not built.

Category Types

Types are inherited from the seed category each image came from. Fusion reaches hit@1 of 0.953 on objects (171 test images), 0.898 on scenes (167), 0.966 on abstract concepts (29) and 0.571 on events (133). Events are clearly the hardest type; there the title/description scorer is the best single scorer (hit@1 0.504), and the zero-shot scorer has the best single-scorer recall@5 (0.746). The abstract result rests on only 29 images and should not be over-read.

Demo and Continual Learning

The Gradio demo accepts an image and an optional title, shows up to ten suggestions with scores, and displays the six most similar known images as evidence. Suggestions below a display cutoff (default 0.05) are hidden, so an image that fits no known category gets no suggestion. A reviewer ticks the correct suggestions and saves the review. Accepted images join the kNN and text index immediately and are saved to disk and reloaded at every start, so nothing is forgotten. Rejected suggestions are logged as negatives, and clicking Save twice adds nothing.

A separate script adds accepted reviews to the training split, retrains the head, TF-IDF and fusion, and reports test P@1 before and after. The system advises and the editor decides; nothing is applied automatically.

Limitations and Ethical Design

Limitations. The sample is small and closed-world (41 seeds, 131 categories), while real Commons is open-ended. Stored labels are incomplete, so strict scores undercount. Seed leakage is reduced by hard mode but not removed, and hard mode rests on few images. There is no grouping by uploader beyond the time split and de-duplication. Category names and the text vocabulary are English-centric. Only two levels of ancestors are fetched, and the display cutoff is tuned on few outside images, so it is approximate.

Ethical design. Human-in-the-loop: advice, never auto-applied. Content categories only, with no identification of people. The code and model weights are open, and only openly licensed Commons data is used. Similar images are shown as an explanation for each result. Multilingual support is future work.

Conclusion

We built a scalable and transparent category suggestion system for Wikimedia Commons. A retrieval-based pipeline fuses four complementary scorers (P@1 0.834, R@5 0.918, mAP 0.873 on a strict time-split test set), uses the category graph so editors see specific suggestions, and keeps humans in control. Fusion gives the best overall ranking but is not best everywhere: zero-shot matching is stronger on torso and tail categories, and events remain the hardest type. The kNN component improves from editor feedback without retraining. Future work: more seeds and categories, a fusion that favors rare categories, multilingual titles, and an approximate index at scale.